

budget, and quality constraints at the transition to the development stage. During the concept stage, the SE team also recognized that the concerns of environmental groups would—and should—impact the approach to the construction of the bridge. The owners took a creative approach by inviting the head of a key environmental group to be part of the board of directors.

From the beginning of the development stage, the owners defined comprehensive requirements and provided definition drawings as part of the contract documents to ensure a project result that not only fulfilled the quality requirements on materials and workmanship but also had the envisioned appearance. The contractor was responsible for the detailed design and for delivering a quality-assured product in accordance with the owners' requirements. The following are representative of the requirements levied at the start of the project:

Schedule: Design life, 100 years; construction time, 1996–2000

Railway: Rail load, International Union of Railways (UIC) 71; train speed, 200 km/h

Motorway: Road axle load, 260 kN; vehicle speed, 120 km/h

Ambient environment: Wind speed (10 min), 61 m/s; wave height, 2.5 m; ice thickness, 0.6 m; temperature, $\pm 27^{\circ}\text{C}$

Ship impact: To pylons, 560 MN; to girder, 35 MN

In addition to established requirements, this project crossed national boundaries and was thereby subject to the legislations of each country. Technical requirements were based on the Eurocodes, with project-specific amendments made to suit the national standards of both countries. Special safety regulations were set up for the working conditions, meeting the individual safety standards of Denmark and Sweden.

The railway link introduced yet another challenge. In Denmark, the rail traffic is right-handed, as on roadways, whereas the trains in Sweden pass on

the left-hand side. The connection needed to ensure a logical transition between the two systems, including safety aspects. In addition, the railway power supply differs between the two countries; thus, it was necessary to develop a system that could accommodate power supply for both railway systems and switch between them on the fly.

The design of a major cable-stayed bridge with approach spans for both road and railway traffic involves several disciplines, including, but not limited to, geotechnical engineering, aerodynamics, foundation engineering, wind tunnel tests, design of piers and pylons, design of composite girders, design of cables and anchorages, design of structural monitoring system, ship impact analysis, earthquake analysis, analysis of shrinkage and creep of concrete, ice load analysis, fatigue analysis, pavement design, mechanical systems, electrical systems, comfort analysis for railway passengers, traffic forecast, operation and maintenance aspects, analysis of construction stages, risk analysis for construction and operation, quality management, and environmental studies and monitoring.

Comprehensive risk analyses were carried out in connection with the initial planning studies, including specification of requirements to secure all safety aspects. Important examples of the results of these studies for the Øresund Bridge were as follows:

Navigation span was increased from 330 to 490 m.

The navigation channel was realigned and deepened to reduce ship groundings.

Pier protection islands were introduced to mitigate bridge/ship accidents.

Risks were considered in a systematic way, using contemporary risk analysis methods such as functional safety analysis using fault tree and “what-if” techniques. Three main issues were considered under the design–build contract:

General identification and assessment of construction risks

Ship collision in connection with realignment of navigation channel

Risks in connection with 5-year bridge operation by contractor

A fully quantified risk assessment of the human safety and traffic delay risks was carried out for a comprehensive list of hazards, including fire, explosion, train collisions and derailments, road accidents, ship collisions and groundings, aircraft collisions, environmental loads beyond design basis, and toxic spillages. An example of a consequence of this analysis was the provision of passive fire protection on the tunnel walls and ceilings.

Both Denmark and Sweden are proud of being among the cleanest industrial countries in the world. Their citizens, and therefore the politicians, would not allow for any adverse environmental impact from the construction or operation of a bridge. The Great Belt and Øresund Strait both constitute corridors between the salty Kattegat and the sweeter water of the Baltic Sea. Any reduction in water exchange would reduce the salt content and, therefore, the oxygen content of the Baltic Sea and would alter its ecological balance. The Danish and Swedish authorities decided that the bridge should be designed in such a way that the flow through of water, salt, and oxygen into the Baltic was not affected. This requirement was designated the zero solution. To limit impacts on the local flora and fauna in Øresund during the construction, the Danish and Swedish authorities imposed a restriction that the spillage of seabed material from dredging operations should not exceed 5% of the dredged amounts. The zero solution was obtained by modeling with two different and independent hydrographical models.

In total, 18 million cubic meters of seabed materials were dredged. All dredged materials were reused for reclamation of the artificial peninsula at Kastrup and the artificial island, Peberholm. A comprehensive and intensive monitoring of the environment was performed to ensure and document the fulfillment of all environmental requirements. In their final status report from 2001, the Danish and Swedish authorities concluded that the zero solution as well as all environmental requirements related to the

construction of the link had been fulfilled. Continual monitoring of eel grass and common mussels showed that, after a general but minor decline, populations had recovered by the time the bridge was opened. Overall, the environment paid a low price at both Øresund and the Great Belt because it was given consideration throughout the planning and construction stages of the bridges.

Conclusions This award-winning bridge is the subject of numerous articles and a doctoral thesis, where details of the construction history and collaboration among all the stakeholders are provided (Jensen, 2014; Nissen, 2006; Skanska, 2013). This project provides a clear example of the benefit of a solid concept stage where the management team was able to resist the customer-driven temptation to jump prematurely into the development stage.

6.3 CASE 3: CYBERSECURITY CONSIDERATIONS IN SYSTEMS ENGINEERING—THE STUXNET ATTACK ON A CYBER-PHYSICAL SYSTEM

Background As our world becomes increasingly digital, the issue of cybersecurity is a factor that the SE practitioner needs to consider. Both hardware and software systems are increasingly at risk for disruption or damage caused by threats taking advantage of digital technologies. Stuxnet, a cyber-attack on Iran's nuclear capabilities discovered in 2010, illustrates the need for the SE practitioner to be comprehensive in application of secure design principles and methods for assessing and avoiding vulnerabilities, and rigorous in mitigation of attack potential (Failliere, 2011; Langner, 2012).

This case study discusses a high degree of attack sophistication previously unseen—malware complexity at military-grade performance, nearly no side effects, and pinpoint accuracy. However, though the creation and deployment of Stuxnet were expensive undertakings, the strategy, tactical methods, and code mechanisms became openly available for others to reuse and build upon at much less expense. Cyber-physical system attacks are becoming increasingly prevalent, and SE must consider the implications of cybersecurity to reduce the vulnerabilities.

Iran's Natanz nuclear fuel enrichment plant (FEP) is a military-hardened facility, with a security fence surrounding a complex of buildings, which are in turn each protected by a series of concrete walls. The complex contains several "cascade halls" for the production of enriched uranium in gas centrifuges. This facility was further hardened with a roof of several meters of reinforced concrete and covered with a thick layer of earth.

Each of the cascade halls is a cyber-physical system, with an industrial control system (ICS) of programmable logic controllers (PLCs), computers, an internal network with no connections to the outside world, and capacity for thousands of centrifuges. Though the internal network is isolated from the outside world by an "air gap," possible vulnerabilities still include malicious insider collusion, non-malicious insider insertion of memory devices brought in from the outside, visiting service technicians, and supply chain intervention. It has been suggested that all of these breach vectors may have played a role in the massive centrifuge damage that began occurring in 2009 and continued at least through 2010.

Malware, now known as Stuxnet, was introduced into the ICS of at least one of the cascade halls and managed to take surreptitious control of the centrifuges, causing them to spin periodically and repeatedly at rates damaging to sustained physical operation. The net effect of the attack is still unclear, but at a minimum, it ranged from disruption of the production process up to potential permanent damage to the affected centrifuges.

Approach Many characteristics of Stuxnet were unprecedented and stand as the inflection point that ushered in a new era of system attack methodology and cyber-physical system targeting. Illuminating forensic analysis of the Stuxnet code was conducted by several well-known cybersecurity firms, with detailed postmortems covered in two documents from the Institute for Science and International Security (Albright, et al., 2010, 2011). This analysis is beneficial in expanding the risk landscape that the SE practitioner should consider during design. Below are some concepts that are concerned in the context of Stuxnet:

Knowing what to do (intelligence)—To be successful, a threat has to be able to take advantage of the targeted system(s). It is uncertain how the perpetrators knew what specific devices were employed in what configuration at Natanz; but after the Stuxnet code was analyzed, Natanz was clearly identified as the target. Stuxnet infected many sites other than Natanz, but it would only activate if that site was configured to certain specific system specifications. The perpetrators needed specific system configuration information to know how to cause damage and also to know how to single out the target among many similar but not identical facilities elsewhere. The SE practitioner needs to consider that adversaries will attempt to gain intelligence on a system and must consider methods to prevent this.

Crafting the code—A zero-day attack is one that exploits a previously unknown vulnerability in a computer application, one that developers have had no time to address and patch. Stuxnet attacked Windows systems outside the FEP using a variety of zero-day exploits and stolen certificates to get proper insertion into the operating system and then initiated a multistage propagation mechanism that started with Universal Serial Bus (USB) removable media infected outside the FEP and ended with code insertion into the ICS inside the FEP. SE practitioners need to be prepared for many different attack vectors (including internal threats) and must consider them during system design.

Jumping the air gap—It was widely believed that Stuxnet crossed the air gap on a USB removable media device, which had been originally infected on a computer outside of the FEP and carried inside. But it was also suggested that the supply chain for PLCs and PLC maintenance personnel may have been at least two additional attack vectors. Whatever the methods, the air gap was crossed multiple times. USB removable media could have also affected a bidirectional transfer of information, sending out detailed intelligence about device types connected to the FEP network subsequently relayed to remote servers outside of the control of the facility. The SE practitioner always needs to

remember that threats to the system are both inside and outside the system boundary.

Dynamic updating—Analysis shows that the attack code, once inserted, could be updated and changed over time, perhaps to take advantage of new knowledge or to implement new objectives. Stuxnet appears to have been continuously updated, with new operational parameters reintroduced as new air gap crossings occur. The SE practitioner needs to prepare for situations after a successful attack has occurred.

Conclusions As the complexity and technology of systems change, the SE practitioner's perspective needs to adjust accordingly. The increasing use of digital-based technologies in system design offers enormous benefits to everyone. However, the introduction of digital technologies also brings different risks than previously dealt with by SE. The case study earlier illustrates a point in time behind us, and the adversarial community continues to evolve new methods. The lesson of this case study is that the SE practitioner needs to understand the threats toward their system(s), be cognizant that attacks can and will occur, and be proactive in protecting their system(s). Robust and dynamic system security needs full engagement of SE (see [Section 3.1.12](#)). A database that the SE practitioners should be aware of is maintained by the National Institute of Standards and Technology (NIST, 2012).

6.4 CASE 4: DESIGN FOR MAINTAINABILITY— INCUBATORS

Note: This case study is excerpted from “Where Good Ideas Come From: The Natural History of Innovation” (Johnson, 2010).

Background In the late 1870s, a Parisian obstetrician named Stephane Tarnier was visiting the Paris Zoo where they had farm animals. While there, he conceived the idea of adapting a chicken incubator to use for human newborns, and he hired “the zoo’s poultry raiser to construct a device that would perform a similar function for human newborns.” At the time, infant

mortality was staggeringly high “even in a city as sophisticated as Paris. One in five babies died before learning to crawl, and the odds were far worse for premature babies born with low birth weights.” Tarnier installed his incubator for newborns at Maternité de Paris and embarked on a quick study of 500 babies. “The results shocked the Parisian medical establishment: while 66 percent of low-weight babies died within weeks of birth, only 38 percent died if they were housed in Tarnier’s incubating box.... Tarnier’s statistical analysis gave newborn incubation the push that it needed: within a few years the Paris municipal board required that incubators be installed in all the city’s maternity hospitals.”

“Modern incubators, supplemented with high-oxygen therapy and other advances, became standard equipment in all American hospitals after the end of World War II, triggering a spectacular 75 percent decline in infant mortality rates between 1950 and 1998.”... “In the developing world, however, the infant mortality story remains bleak. Whereas infant deaths are below ten per thousand births throughout Europe and the United States, over a hundred infants die per thousand (births) in countries like Liberia and Ethiopia, many of them premature babies that would have survived with access to incubators. But modern incubators are complex, expensive things. A standard incubator in an American hospital might cost more than \$40,000 (about €30,000). But the expense is arguably the smaller hurdle to overcome. Complex equipment breaks, and when it breaks, you need the technical expertise to fix it. You also need replacement parts. In the year that followed the 2004 Indian Ocean tsunami, the Indonesian city of Meulaboh received eight incubators from a range of international relief organizations. By late 2008, when an MIT professor named Timothy Prestero visited the hospital, all eight were out of order, the victims of power surges and tropical humidity, along with the hospital staff’s inability to read the English repair manual. The Meulaboh incubators were a representative sample: some studies suggest that as much as 95% of medical technology donated to developing countries breaks within the first 5 years of use.”

Approach “Prestero had a vested interest in those broken incubators, because the organization he founded, Design that Matters, had been working for several years on a scheme for a more reliable, and less expensive, incubator,

one that recognized complex medical technology was likely to have a very different tenure in a developing world context than it would in an American or European hospital. Designing an incubator for a developing country wasn't just a matter of creating something that worked; it was also a matter of designing something that would break in a non-catastrophic way. You couldn't guarantee a steady supply of spare parts, or trained repair technicians. So instead, Prestero and his team decided to build an incubator out of parts that were already abundant in the developing world. The idea had originated with a Boston doctor named Jonathan Rosen, who had observed that even the smaller towns of the developing world seemed to be able to keep automobiles in working order. The towns might lack air conditioning and laptops and cable television, but they managed to keep their Toyota 4Runners on the road. So, Rosen approached Prestero with an idea: What if you made an incubator out of automobile parts?"

"Three years after Rosen suggested the idea, the Design that Matters team introduced a prototype device called NeoNurture. From the outside, it looked like a streamlined modern incubator, but its guts were automotive. Sealed-beam headlights supplied the crucial warmth; dashboard fans provided filtered air circulation; door chimes sounded alarms. You could power the device via an adapted cigarette lighter, or a standard-issue motorcycle battery. Building the NeoNurture out of car parts was doubly efficient, because it tapped both the local supply of parts themselves and the local knowledge of automobile repair. These were both abundant resources in the developing world context, as Rosen liked to say. You didn't have to be a trained medical technician to fix the NeoNurture; you didn't even have to read the manual. You just needed to know how to replace a broken headlight."

Conclusions Sometime the highest technology solution is not the best. SE practitioners need to consider issues like maintainability and logistics at the project outset in the concept and development stages. It is too late to address these in later stages.

6.5 CASE 5: ARTIFICIAL INTELLIGENCE IN SYSTEMS ENGINEERING—AUTONOMOUS VEHICLES

Note: Much of the information in this case study is derived from the United States National Transportation Safety Board (NTSB) report on automation (2019a). Page numbers are indicated.

Background On March 18, 2018, a pedestrian walking a bicycle was fatally struck by a 2017 Volvo XC90 Uber vehicle operating an Automated Driving System (ADS) then under development by Uber's Advanced Technologies Group (ATG). The Volvo's advanced driver assistance system was disabled to prevent conflicts with its radar which operated on the same frequency as the radar for Uber's ATG ADS (p. 15).

At the time of the pedestrian fatality, the ATG-ADS had used one lidar and eight radars to measure distance; several cameras for detecting vehicles, pedestrians, reading traffic lights, and classifying detected objects; and various sensors that had been recently calibrated for telemetry, positioning, monitoring of people and objects, communication, acceleration, and angular rates. It also had a human-machine interface (HMI) tablet and a Global Positioning System (GPS) used solely to assure that the car was on an approved and pre-mapped route before engaging the ADS. The ADS allowed the vehicle to operate at a maximum speed of 45 mph (p. 7), to travel only on urban and rural roads, and under all lighting and weather conditions except for snow accumulation. The ADS system was easily disengaged; until then, almost all of its data was recorded, with the exception noted below of lost data occurred whenever an alternative determination of an object was made by ADS (e.g., shifting from an "object" in the road to an oncoming "vehicle" ahead).

Approach Designing the interactions of a human and a machine to form into a team (or system) that also acts autonomously requires significant shifts in thinking, modeling, and practice. This begins with changing the unit of analysis from individual humans or programmable machines to teams.

The ADS constructed a virtual environment from the objects that its sensors detected, tracked, classified, and then prioritized based on fusion processes (p. 8). ADS predicted and detected any perceived object's goals and paths as part of its classification system. However, if classifications were made and then changed, as happened in this case (e.g., from "object" to "vehicle" and back to "object"), the prior tracking history was discarded. A flaw since corrected. Also, pedestrians outside of a crosswalk were not assigned a predicted track. Another flaw since corrected.

When ADS detected an emergency (p. 9), it suppressed taking any action for one second to avoid false alarms. After the one second delay, the car's self-braking or evasion could begin. Another major flaw since corrected (p. 15). If a collision could not have been avoided, an auditory warning was to be given to the operator at the same time that the vehicle was to be slowed (in this case, the vehicle may have also begun to slow because an intersection was being approached).

As shown in [Figure 6.1](#), using the recorded data to replay the accident, before impact: Radar first detected the pedestrian 5.6 seconds; Lidar made its first detection at 5.2 seconds, classified the object as unknown and static, changed to a static vehicle at 4.2 seconds on a path predicted to be a miss, reclassified to "other" and static but back again to vehicle between 3.7 to 2.7 seconds, each re-classification discarding its previous prediction history for that object; then a bicycle, but static and a miss at 2.6 seconds; then unknown, static and a miss at 1.5 seconds; then classified the object a bicycle and an unavoidable hazard at 1.2 seconds, the categorization of a hazard immediately initiating "action suppression;" after the 1 seconds pause, finally an auditory alert was sounded at 0.2 seconds; the operator took control at 0.02 seconds before impact; and the operator selected brakes at 0.7 seconds after impact.

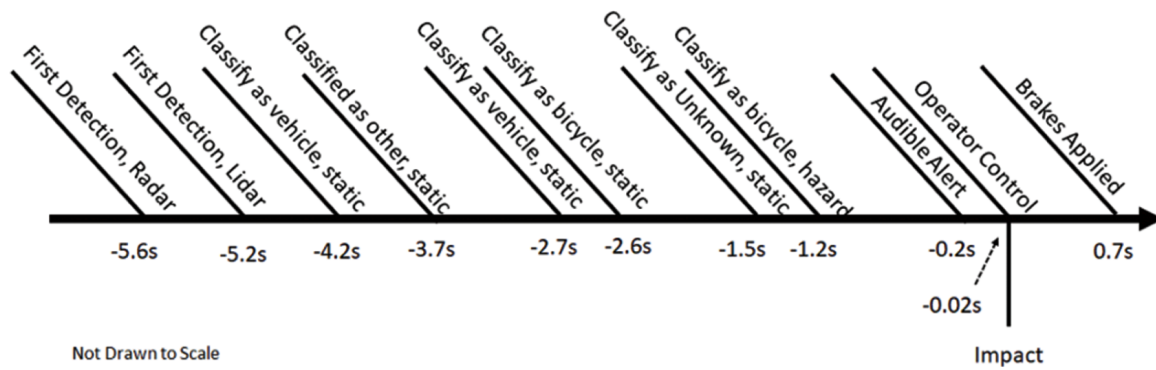


FIGURE 6.1 Timeline of vehicle impact. From NTSB (2019a). used with permission. All other rights reserved.

The ADS failed to correctly predict the detected object's path, and only determined it to be a hazard at 1.2 seconds before impact, causing any action to be suppressed for one second but, and as a consequence of the impact anticipated in the shortened time-interval remaining before impact, exceeding the ADS design specifications for braking, and thus not enacted; after this self-imposed one second delay, an auditory alert was sounded (p. 12). For almost 20 minutes before impact, the HMI presented no requests for its human operator's input (p. 13), likely contributing to the human operator's sense of complacency.

Conclusions The following lessons can be taken from this case study:

- The operator was distracted by their personal cell phone; the pedestrian's blood indicated that they were impaired from drugs and that they violated Arizona State's policy by jaywalking.
- The indecisiveness of the ADS was partly attributed to the pedestrian not being in a crosswalk, a feature the system was not designed to address (p. 12), since corrected.
- Uber had inadequate safety risk assessments of its procedures, ineffective oversight in real-time of its vehicle operators to determine whether they were being complacent and exhibited overall an inadequate safety culture (p. vi; see also NTSB, 2019b).
- The Uber ADS was functionally limited, unable to correctly classify the object as a pedestrian, to predict their path, or to adequately assess its risk until almost impact.
- The ADS's decision to suppress action for one second to avoid false alarms increased the risk of driving on the roads and prevented the brakes from

being applied immediately to avoid a hazardous situation.

- By disconnecting the Volvo car's own safety systems, Uber increased risk by eliminating the redundant safety systems for its ADS, since corrected (p. vii).

According to NTSB's decision, although the National Highway Traffic Safety Administration (NHTSA) had published a third version of its automated vehicles policy, NHTSA provided no means to a self-driving company of evaluating its vehicle's ADS to meet national or State safety regulations, or to provide a company with the detailed guidance to design an adequate ADS to operate safely. NTSB recommended that safety assessment reports submitted to NHTSA, voluntary at the time of NTSB's final report, be made mandatory (p. viii) and uniform across all states (e.g., Arizona had taken no action by the time of NTSB's final report).

At its most basic level, this case study explores the fundamental tool of debate used for eons by autonomous humans confronting uncertainty. It concludes that machines using artificial intelligence (AI) to operate as members of a team must be able to tell its human partners whenever the machines perceive a change in the context that affects their team's performance; in turn, AI machines must be able to understand humans conversing with them, their interactions indicating the team's intelligence (Cooke, 2020). Moreover, as part of a team, once these AI governed machines learn what humans want them to learn, they will know when the human members of their team are either complacent or malicious in the human's performance of their roles (Lawless, et al., 2017), a capability not yet available in real time, but possibly over the next five years (Sofge, et al., 2019).

From a human-machine team's perspective, the Uber car was a poor team player (Lawless, 2019). Human teams are autonomous. The operator and vehicle operated independently of each other. Facing uncertain situations, the NTSB report confirmed that no single human or machine agent can determine context alone. Resolving uncertainty requires at a minimum a state of shared interdependence to build context, to adapt to rapid changes in context, and, overall, to operate safely and ethically autonomous human-machine systems. We also know from Cummings (2015) that the best science

teams are fully interdependent. Cooke (2020) locates a team's intelligence in the interdependent interactions among its teammates. And to reduce uncertainty in an autonomous system necessitates that human and machine teammates are able to explain to each other, however imperfectly, their view of reality in causal terms (Pearl, 2002; Pearl and Mackenzie, 2018).

As SE moves into the future with the design and operation of autonomous human-machine teams and systems, an interdependence among the parts of a team not only makes the parts reactive to the presence of each other, but it also allows for the team to emerge as a single unit, increasing the team's performance. SE practitioners must be cognizant of the benefits and downfalls of AI and autonomy.

6.6 OTHER CASE STUDIES

Additional case studies can be found in other sources. The SEBoK maintains a set of case studies identified as "implementation examples"

(https://www.sebokwiki.org/wiki/Systems_Engineering_Implementation_Examples).

The SEBoK case studies span domains including defense systems, information systems, management systems, medical systems, space systems, transportation systems, and utility systems.

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APPENDIX B: ACRONYMS

Note: Other acronyms may be defined as used under their respective appendices. Abbreviations used for references are described in Appendix A. Abbreviations used for the system life cycle processes are described in Appendix D.

A _a	Achieved availability
A _i	Inherent availability
A _o	Operational availability
ADS	Automated Driving Systems
AECL	Atomic Energy Commission Limited [Canada]
AF	Architecture Framework
AI	Artificial Intelligence
AIAA	American Institute of Aeronautics and Astronautics [United States]
ALARP	As low as reasonably practicable
ALT	Accelerated Life Testing
ANSI	American National Standards Institute [United States]
API	American Petroleum Institute
API	Application programming interface
ARAP	As Resilient as Practicable
ARP	Aerospace Recommended Practice
ASEP	Associate Systems Engineering Professional [INCOSE]
ASOT	Authoritative Source of Truth

ASQ	American Society for Quality
ATAM	Architecture tradeoff analysis method
AWG	Automotive Working Group [INCOSE]
BIT	Built-In Test
CAD	Computer-aided design
CAIV	Cost as an independent variable
CBA	Cost–benefit analysis
CBS	Cost Breakdown Structure
CCB	Configuration Control Board
CE	Conformité Européenne [EU]
CE	Cost Effectiveness
CEA	Cost-effectiveness analysis
CFD	Computational Fluid Dynamics
CI	Configuration Item
CI/CD	Continuous Integration/Continuous Delivery
CM	Configuration Management
CMMI®	Capability Maturity Model® Integration [CMMI Institute]

COCOMO	Constructive Cost Model
ConOps	Concept of operations
COSYSMO	Constructive Systems Engineering Cost Model
COTS	Commercial off-the-shelf
CPS	Cyber-physical system
CRB	Configuration Review Board
CSEP	Certified Systems Engineering Professional [INCOSE]
DANSE	Designing for Adaptability and evolutionN in System of Systems Engineering
DAU	Defense Acquisition University [United States]
DD	Data Dictionary
DE	Digital Engineering
DevOps	Development, Operations
DevSecOps	Development, Security, Operations
DFD	Data flow diagrams
DFM	Design For Manufacturing
DFT	Design For Testability
DFX	Design For X
DMSMS	Diminishing manufacturing sources and material shortages
DoD	Department of Defense [United States]
DoDAF	Department of Defense Architecture Framework [United States]

DSM	Design Structure Matrix
DT	Design Thinking
DTC	Design to cost
EIA	Electronic Industries Alliance
EPD	Environmental Product Declaration
ESEP	Expert Systems Engineering Professional [INCOSE]
EU	European Union
FAA	Federal Aviation Administration [United States]
FBS	Functional Breakdown Structure
FCA	Functional configuration audit
FD/FI	Failure detection/Failure isolation or Fault Detection/Fault Isolation
FEA	Finite Element Analysis
FEP	Fuel enrichment plant
FFBD	Functional flow block diagram
FMEA	Failure Mode and Effects Analysis
FMECA	Failure modes, effects, and criticality analysis
FTA	Fault tree analysis
FuSE	Future of Systems Engineering [INCOSE]

G&A	General and administrative
GAO	Government Accountability Office [United States]
GEIA	Government Electronics & Information Technology Association
GERAM	Generalized Enterprise Reference Architecture and Methodology
GNP	Gross national product
GPS	Global Positioning System
GRIP	Governance for Railway Investment Projects
HALT	Highly accelerated life testing
HCD	Human-Centered Design
HFE	Human factors engineering
HITL	Human-in-the-loop
HMI	Human Machine Interface
HPC	High Performance Computing
HSI	Human systems integration
IAD	Interface Agreement Document
IC	Initial cost
ICD	Interface Control Document
ICS	Industrial control system

ICSM	Incremental Commitment Spiral Model
ICWG	Interface Control Working Group
IDD	Interface Definition Document
IE	Industrial engineering
IEC	International Electrotechnical Commission
IEEE	The IEEE [formerly the Institute of Electrical and
IIESE	Electronics Engineers] Institute of Industrial and Systems Engineers
ILS	Integrated logistics support
INCOSE	International Council on Systems Engineering
INFORMS	Institute for Operations Research and the Management
IoT	Sciences Internet of Things
IPAL	INCOSE Product Asset Library [INCOSE]
IPDT	Integrated Product Development Team
IPO	Input–process–output
IPT	Integrated Product Team
ISEBoK	Industrial and SE Body of Knowledge
ISO	International Organization for Standardization
IT	Information technology
ITS	Intelligent Transportation System
JAXA	Japan Aerospace Exploration Agency [Japan]
JERG	JAXA Engineering Requirement, Guideline [Japan]

KM	Knowledge management
KPP	Key Performance Parameter
LAI	Lean Advancement Initiative
LCA	Life cycle assessment
LCC	Life cycle cost
LCIA	Life cycle impact assessment
LCM	Life cycle management
LCO	Life cycle optimization
LDSE	Loss-Driven Systems Engineering
LEfMEP	Lean Enablers for Managing Engineering Programs
LINAC	Linear accelerator
LORA	Level of Repair Analysis
MA&S	Modeling, analysis, and simulation
MaaS	Mobility as a Service
MBSE	Model-based systems engineering
MFESA	Method Framework for Engineering System Architectures
MIT	Massachusetts Institute of Technology [USA]
ML	Machine Learning
MODA	Multiple objective decision approach

MOE	Measure of effectiveness
MOP	Measure of performance
MTBF	Mean time between failure
MTTR	Mean time to repair
MVP	Minimum viable product
N ²	N-squared diagram
NAF	NATO Architecture Framework
NAFEMS	The International Association for the Engineering Modelling, Analysis and Simulation Community
NASA	National Aeronautics and Space Administration [United States]
NCOSE	National Council on Systems Engineering (INCOSE, pre-1995)
NCS	Network-Centric Systems
NDI	Non-developmental item
NDIA	National Defense Industrial Association [United States]
NIH	Not Invented Here
NIST	National Institute of Standards and Technology [United States]
O&G	Oil and Gas
OBS	Organizational Breakdown Structure

OEM	Original Equipment Manufacturer
OMG	Object Management Group
OOSEM	Object-Oriented Systems Engineering Method
OpsCon	Operational concept
OR	Operations Research
PBS	Product Breakdown Structure
PCA	Physical configuration audit
PEAF	Pragmatic Enterprise Architecture Framework
PESTEL	Political, Economic, Social, Technological, Environmental, and Legal
PHS&T	Packaging, handling, storage, and transportation
PLC	Programmable logic controller
PLE	Product line engineering
PLM	Product line management
PMBok	Project Management Body of Knowledge [PMI]
PMI	Project Management Institute
PMP	Project Management Plan
PPP	Public–Private Partnership
PSM	Practical Software and Systems Measurement
QA	Quality assurance
QC	Quality characteristics

QC	Quality control
QM	Quality management
R&D	Research and development
RAM	Reliability, availability, and maintainability
RBD	Reliability block diagram
RCM	Reliability-centered maintenance
RFP	Request for proposal
RFQ	Request for quotation
RMP	Risk management plan
ROI	Return on investment
SAE	SAE International [formerly the Society of Automotive Engineers]
SAFe	Scaled agile framework
SaMDs	Software as Medical Devices
SBD	Set-Based Design
SC	Sustainment cost
SCF	Situation Context Framework
SCM	Supply chain management
SE	System effectiveness

SE	Systems engineering
SEBoK	Guide to the Systems Engineering Body of Knowledge
SECAG	Systems Engineering Competency Assessment Guide [INCOSE]
SECF	Systems Engineering Competency Framework [INCOSE]
SEH	Systems Engineering Handbook [INCOSE]
SEIPS	Systems Engineering Intervention for Patient Safety
SEIT	Systems Engineering and Integration Team
SEMP	Systems Engineering Management Plan
SEMS	Systems Engineering Master Schedule
SEP	Systems Engineering Plan
SEQM	System Engineering Quality Management
SLA	Service-level agreement
SMSWG	Systems Modeling and Simulation Working Group [NAFEMS and INCOSE]
SoI	System of interest
SoS	System of systems
SOW	Statement of work
SPC	Statistical Process Control
SROI	Social return on investment

SSE	System Security Engineering
STEM	Science, technology, engineering, and mathematics
STPA	System-theoretic process analysis
SWaP	Size, weight, and power
SWE	Software engineering
SWOT	Strengths, Weaknesses, Opportunities, Threats
SysML™	Systems Modeling Language [OMG]
TADSS	Training Aids, Devices, Simulators, and Simulations
TCO	Total cost of ownership
TOC	Total ownership cost
TOGAF	The Open Group Architecture Framework [The Open Group]
TOP	Technology, Organization, People
TOWS	Threats, Opportunities, Weaknesses, and Strengths
TP	Transaction-processing
TPM	Technical performance measure
TLI	Technical Leadership Institute [INCOSE]
TR	Technical report
TRL	Technology readiness level
UAF	Unified Architecture Framework

UAM	Unified Architecture Method
UI	User Interface
UIC	International Union of Railways
UK	United Kingdom
UL	Underwriters Laboratory [United States and Canada]
US/USA	United States/ United States of America
USB	Universal Serial Bus
USD	US dollars [United States]
UX	User Experience or User eXperience
V&V	Verification and Validation or Verify and Validate
VSE	Very Small Entities or Very Small Enterprises
VV&A	Verification, validation, and accreditation
WBS	Work Breakdown Structure
WG	Working group
WLC	Whole Life Cost
WP	Work package
WPA	Work Process Analysis
XP	Extreme Programming
ZD	Zero Defect
ZDA	Zero Defect Attitude

APPENDIX C: TERMS AND DEFINITIONS

Note: Terms that carry meanings consistent with general dictionary definitions are not included in this glossary. Sources of definitions are as indicated. Other related terms can be found in ISO/IEC/IEEE 24765 (2017) and SEVOCAB (2023). Definitions of the typical inputs and outputs on the IPO diagrams can be found in Appendix E.